Same Demand, Different Arrival Times: Parking-Search Congestion in an Agent-Based Model

[Zachary Wang 24648002] | [Maneesh Devireddy 25204945]

1. Background and research aims

As we know, parking demand can be concentrated around a shared event, such as the start of a class: two periods may have the same number of arriving vehicles but different arrival patterns and a short burst may create a queue that takes time to clear. We investigate this process in a simplified campus-style car park. The campus setting motivates the question; however, the model doesn't represent a particular university car park.

Parking search has been studied as a source of additional traffic. Shoup (2006) models drivers' choice between searching for cheaper curb parking and paying for immediately available off-street parking. This is about parking search to prices and the value of time. Benenson, Martens and Birfir (2008) present 'PARKAGENT,' which represents individual drivers in a spatial environment. Their Tel Aviv case study examines additional parking supply.

These studies establish parking as a problem involving individual decisions and shared resources. Our smaller model keeps parking supply fixed and examines arrival timing. It doesn't include price choices, walking distances or a real street network.

Our research question is: under fixed total vehicle demand, how does arrival synchronisation affect waiting times and backlog, and how does this effect vary with parking turnover? We expect shorter arrival windows to increase waiting and peak backlog. We also examine whether slower turnover makes the additional delay from concentrated arrivals larger. These are separate questions: longer parking stays could increase waiting in both arrival scenarios without increasing the difference between them.

The project's contribution is a repeated comparison that separates arrival concentration from total demand. It combines competition for parking spaces, temporary blocking during parking manoeuvres, and an outside queue. We measure vehicles waiting outside as well as vehicles searching inside. This makes the investigation distinct from a general road-traffic demonstration: the question is about access to finite parking resources and the timing of demand. Although the main analysis does not use entry control, the implementation supports it as a separate extension.

2. Model design

Each vehicle is an agent with a state, location, parking timer and event timestamps. The environment is a one-way circular road with 30 discrete positions and 12 parking spaces placed approximately evenly around it. Besides that, at most one vehicle occupies each road position or parking space. The entrance is at road position 0. We created this small layout to make vehicle interactions easy to inspect and keeps the experiments inexpensive.

Vehicles pass through waiting, searching, parking, parked and departed states. An outside FIFO queue preserves arrival order. Searching vehicles move one road position per update when the movement rule allows. When a vehicle finds an adjacent empty space, it starts a two-step parking manoeuvre. It remains on the road during the manoeuvre and can block vehicles following it. These local interactions produce the system-level backlog.

Movement is updated simultaneously. Adjacent searching vehicles can move together when the road ahead is free to move. A vehicle performing a parking manoeuvre is stationary, and its blockage propagates backwards through adjacent searching vehicles, including across the loop boundary. Parked vehicles leave directly when their timers expire; they do not rejoin the road and the entrance admits at most one vehicle per step and only when its road position is free.

Initially, nine of the twelve spaces are unavailable, with locations selected randomly. Initial occupants and newly parked vehicles receive positive parking durations sampled as $\max\left(1, \text{round}\left(\text{Exponential}\left(d\right)\right)\right)$. The parameter d controls parking turnover. Rounding means the discrete mean is only approximately d. This simple distribution introduces variation while keeping the model manageable; it has not been fitted to campus parking data.

Scheduled arrivals join the outside queue before each step. The update then processes parked departures, progresses existing parking manoeuvres, starts new manoeuvres, moves searching vehicles, and admits the first waiting vehicle if possible. Finally, the model records the state. Random automatic arrivals are disabled in the main experiment. This order is unchanged across all conditions.

3. Experimental design and verification

We vary arrival-window width w over 10, 20, 40 and 60 steps; total arrivals N over 12, 18 and 24 vehicles; and parking-duration parameter d over 15, 30 and 60 steps. These ranges compare concentrated and spread arrivals, demand equal to or above parking capacity, and faster and slower turnover. Within each width comparison, all other settings remain fixed. Each condition uses seeds 0–9 and runs for 300 steps, giving 36 conditions and 360 runs.

For vehicle i = 0, …, N − 1, its scheduled arrival is $40 - \lfloor \frac{w}{2} \rfloor + \text{round}\left[\frac{i\left(w - 1\right)}{N - 1}\right]$. This keeps the event approximately centred around step 40. For example, width 10 uses steps 35–44, while width 60 uses steps 10–69. Multiple arrivals can occur in one step. The same seed set is reused, but different event sequences can consume random draws differently, so corresponding vehicles don’t need to get identical parking durations.

Waiting time is parking-completion time minus arrival time, including outside waiting, searching and the manoeuvre. Its mean is only vehicles that successfully park. Completion fraction and unfinished-parking count are therefore reported with it. Completion means reaching a parked state, not departing. Backlog B(t) is searching-inside plus waiting-outside vehicles; it excludes manoeuvres. We report max B(t) and cumulative backlog, the sum of B(t) across steps, in vehicle-steps. Means and population standard deviations summarise runs; error bars are descriptive, not confidence intervals.

The supplied version successfully passes 28 automated tests covering movement, blocking, parking, entry control, reproducibility, completion and animation consistency. They also check movement across the loop boundary, FIFO admission, vehicle conservation, and departure before admission. The animation reuses the experiment's arrival and stepping functions. Identical settings produce matching model records. Repeated or earlier target steps leave the model state unchanged. These checks verify implementation behaviour; they do not validate the model against real parking data.

4. Results

Across all nine combinations of total demand and parking duration, mean waiting time decreased as the arrival window widened from 10 to 60 steps. All scheduled vehicles successfully parked within 300 steps in all 360 runs. Thus, the waiting-time means in this sweep exclude no unfinished scheduled vehicles.

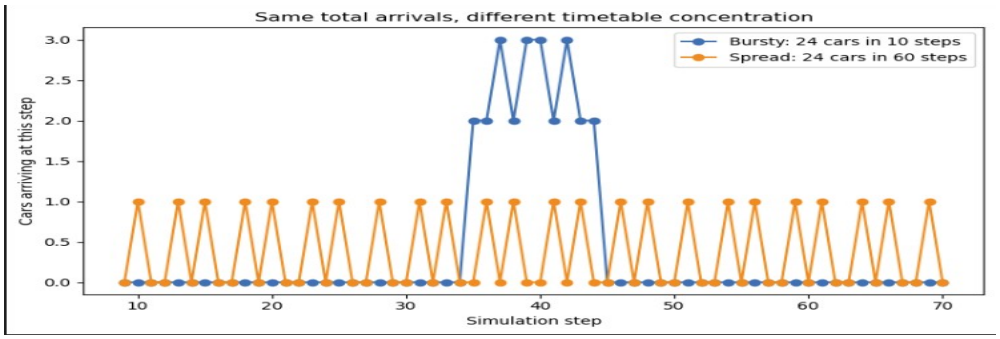


Figure 1. Two schedules containing 24 arrivals each

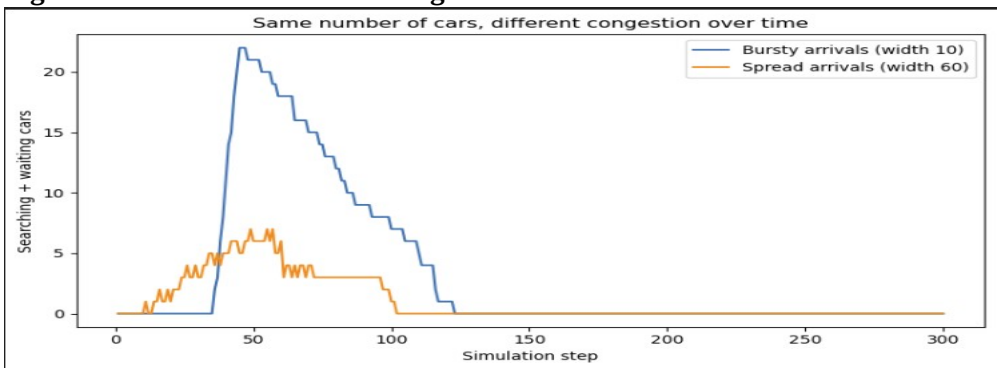


Figure 2. Backlog trajectories for 24 arrivals with d = 30 and seed 3 over 300 simulation steps.

The example in Figure 2 shows a rapid backlog build-up under concentrated arrivals, followed by a decrease period. Spread arrivals produce a smaller peak. Both curves return to zero, so this example is just a temporary congestion. In the following, Table 1 summarizes the corresponding repeated experiments and the other parking-duration settings.

Table 1. Results for 24 arrivals, averaged across ten seeds. Waiting is mean ± standard deviation in steps. Peak backlog is in vehicles; cumulative backlog is in vehicle-steps.

d	Width	Waiting	Peak backlog	Cumulative
15	10	32.57 ± 2.92	21.70	733.60
15	60	11.63 ± 2.07	5.80	231.00
30	10	46.04 ± 7.17	22.00	1056.90
30	60	18.83 ± 3.75	8.20	403.80
60	10	60.99 ± 10.91	22.40	1415.70
60	60	42.90 ± 14.63	12.20	981.60

As we can see, for N = 24 and d = 30, widening the arrival window from 10 to 60 reduced average waiting from 46.04 to 18.83 steps, about 59%. Mean peak backlog fell from 22.00 to 8.20 vehicles, while cumulative backlog fell from 1056.90 to 403.80 vehicle-steps. The change therefore affected both the peak and the amount of congestion accumulated over time.

5. Discussion

The results support the hypothesis that concentrating a fixed demand increases congestion under the tested conditions. However, the effect depends on the outcome being compared. Longer parking stays increased waiting in both arrival scenarios. However, the difference in waiting time between concentrated and spread arrivals didn't consistently increase as parking duration increased.

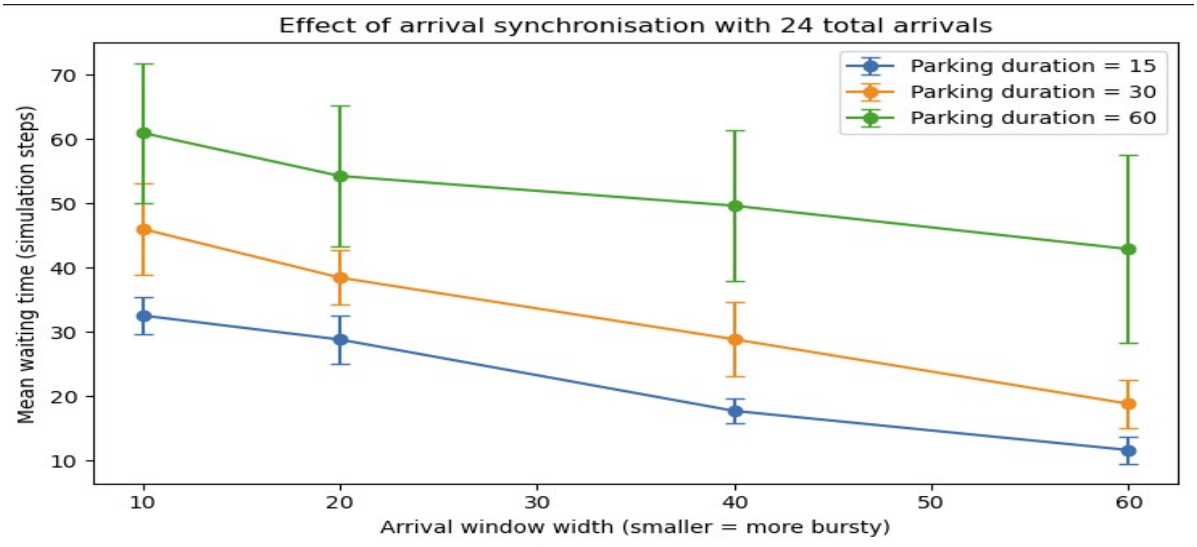


Figure 3. Mean waiting time for 24 arrivals across the four window widths. Each point represents ten seeds; error bars show ±1 population standard deviation. Duration denotes the exponential parking-duration parameter d, in simulation steps.

For 24 arrivals, the difference between widths 10 and 60 was 20.94 steps at d = 15, 27.21 at d = 30, and 18.09 at d = 60, calculated before rounding. The middle setting had the largest difference. At d = 60, even spread arrivals waited 42.90 steps on average. A possible explanation is that slow turnover leaves vehicles competing for occupied spaces in both scenarios, reducing the advantage of spreading arrivals. The present experiments don't isolate its cause.

The entrance is also an important point. Twenty-four arrivals over ten steps exceed the maximum admission rate of one vehicle per step. An outside queue can therefore form even if some spaces are available. For N = 24 and d = 30, the mean peak outside queue was 17.40 vehicles for width 10 and 0.50 for width 60. Parking manoeuvres can further block access and movement. We therefore regard the timing effect as a combined result of admission capacity, parking availability and road interactions, not as an effect of parking-space shortage only.

Higher demand also increased waiting. At d = 30 and width 10, mean waiting increases from 24.93 steps for 12 arrivals to 37.09 for 18 and 46.04 for 24. Keeping demand fixed within each width comparison was therefore necessary. Otherwise, a larger queue could simply reflect more arriving vehicles.

In practical terms, arrival scheduling also deserves attention. The model shows that spreading a peak can help without adding spaces, but the size of that benefit depends on the wider system. It does not establish that a particular timetable change would get the same reduction at a real campus.

6. Limitations and reproducibility

The model uses a simple loop, identical movement rules, and no alternative destinations. Departures disappear from their spaces, avoiding exit conflicts. Parking durations follow an assumed distribution, and simulation steps have no calibrated duration in minutes. These choices help explain the model's behaviour, but they limit transfer to real parking systems. As a result, the result is only about this simplified system.

Initial conditions require additional attention. The car park starts 75% occupied, but initial vehicles can leave before later arrivals. Changing d affects both those departures and the stays of newly parked vehicles. The common-centre arrival schedules are therefore compared within a transient system, not at unchangeable occupancy. Useful follow-up experiments would vary initial occupancy, separate the initial vehicles' remaining stays from new parking durations, and test another duration distribution. Further seeds would help assess the stability of the observed differences.

The Python 3 implementation, requirements and CSV outputs support reproduction. From the project root, run python -m src.experiments for the main sweep, python -m pytest -q for tests, and python -m src.animate_model for the live comparison. The notebook notebooks/checkpoint2.ipynb presents the analysis. Code and simulation instructions are available at https://github.com/ZacAster/CITS4403_Project

The report uses the fixed-demand sweep; historical random-arrival data and the optional entry-control experiment are not used as evidence.

7. Conclusion

Arrival timing mattered even when total demand was unchanged. Wider arrival windows reduced mean waiting across all nine tested demand–duration combinations. For 24 arrivals and d = 30, widening the window from 10 to 60 steps reduced mean waiting by about 59%. Slower turnover raised waiting overall, but did not consistently increase the penalty from concentrated arrivals. The model provides a clear, reproducible example of congestion emerging from individual parking actions and shared constraints. Instead of predicting a real world outcome, its main value is explaining these interactions and identifying questions for further testing.

References

[1] Benenson, I., Martens, K., & Birfir, S. (2008). PARKAGENT: An agent-based model of parking in the city. Computers, Environment and Urban Systems, 32(6), 431–439. https://doi.org/10.1016/j.compenvurbsys.2008.09.011

[2] Shoup, D. C. (2006). Cruising for parking. Transport Policy, 13(6), 479–486. https://doi.org/10.1016/j.tranpol.2006.05.005